

Cognitive Models, Wisdom of the Crowd, Citizen Science, and Frogs: An Extension of Brown et al. (2024)

Michael D. Lee

University of California Irvine

Detecting Frogs

- Citizen scientists listen to audio recording clips trying to detect the presence of various ground and tree frogs by their calls

environment



scientist



data



Detecting Frogs

- Multiple citizen scientists can independently listen to the same clip

environment



scientist



data



Detecting Frogs

- The same citizen scientist can listen to multiple clips

environment



scientist

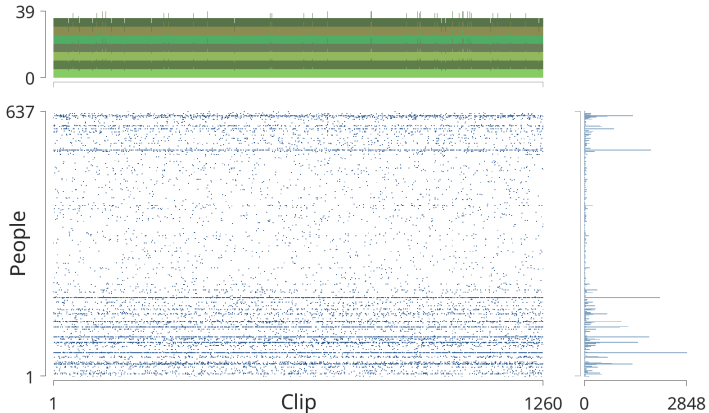


data



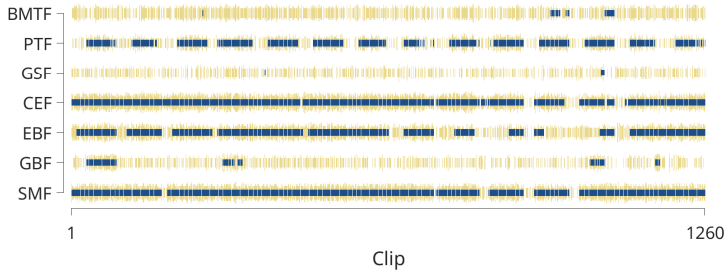
Data

- The joint and marginal distributions of people judging clips shows
 - lots of people judge few clips, but a few people judge many
 - the data collection system aims to have five judgments for each frog in each clip



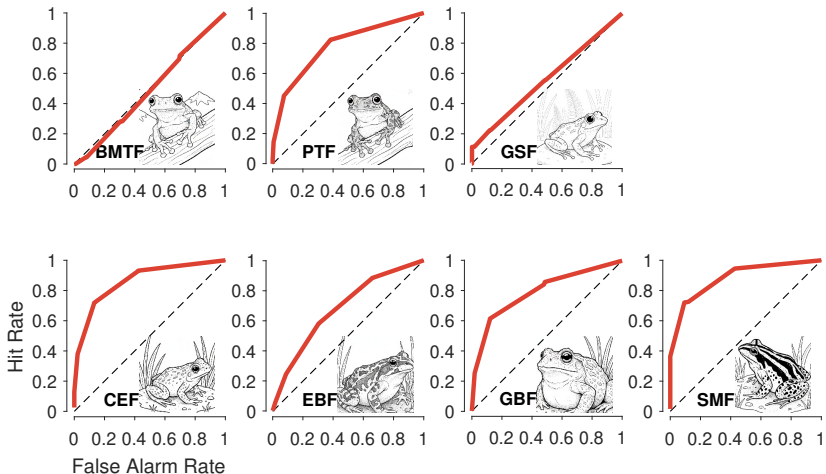
Crowd Proportions and the Truth

- The blue marks indicate the ground truth (expert derived) presence of each type of frog across the sequence of clips
- The yellow bars show the proportion of citizen scientists who judged a frog was present in each clip



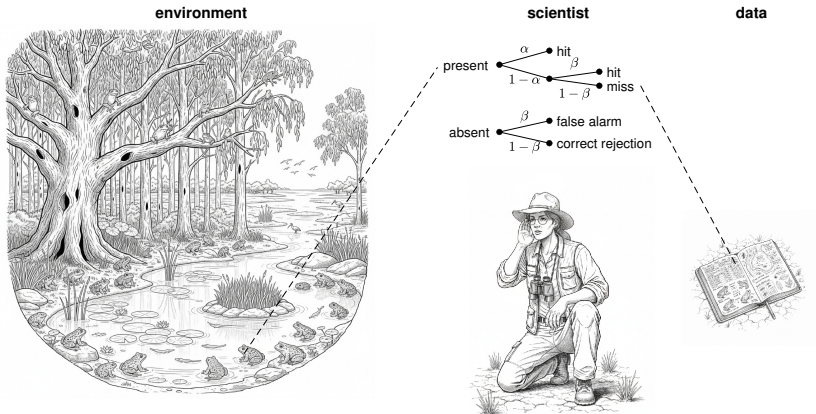
Crowd Performance

- ROC curves show reasonable wisdom the crowd performance for five of the frog types
 - the BMTF and GSF tree frogs are not well detected
 - both are rarely present but people produce many false alarms



Model-Based Wisdom of the Crowd

- The model-based approach uses cognitive models to explain how the behavior data were produced (Brown et al., 2024; Lee, 2024)
 - make inferences about the cognitive abilities and biases of individuals
 - make inferences about latent states of nature



Model

- The model assumes frog f in clip j is either $\tau_{fj} = 1$ present or $\tau_{fj} = 0$ absent with a base-rate that depends on the frog ψ_f

$$\tau_{fj} \sim \text{Bernoulli}(\phi_j)$$

$$\phi_j \sim \text{Uniform}(0, 1)$$

- Individual i judges frog f in clip j to be $y_{ijf} = 1$ present with probability θ_{ijf} given by the one-high threshold model

$$\alpha_i, \beta_i \sim \text{Uniform}(0, 1)$$

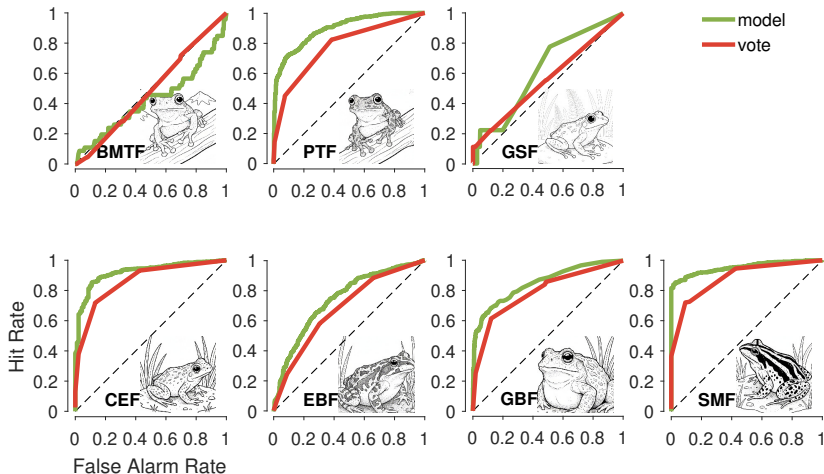
$$\theta_{ijf} = \begin{cases} \alpha_i + (1 - \alpha_i) \beta_i & \text{if } \tau_{fj} = 1 \\ \beta_i & \text{if } \tau_{fj} = 0 \end{cases}$$

$$y_{ij} \sim \text{Bernoulli}(\theta_{ij})$$

- Implemented the model in JAGS and Stan

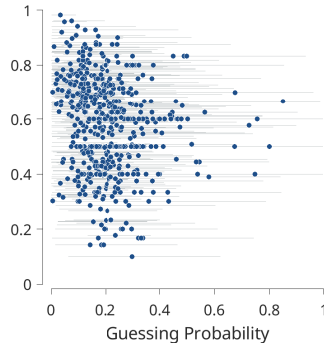
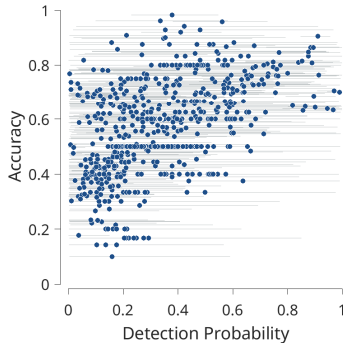
Model-Based Crowd Performance

- The performance of the model closely matches the results for the Brown et al. (2024) model
 - both model-based approaches improve on the tallying judgements



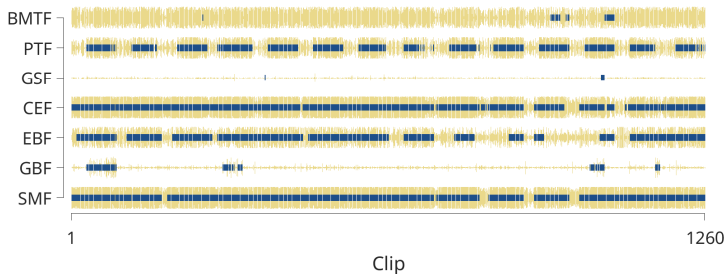
Inferences about Detection and Guessing Bias

- Inferences about detection ability α_i and guessing bias β_i provide some **prediction** about the performance of individual i
 - this allows, for example, “yes” judgments from non-experts to be explained away as guessing



Model-Based Crowd and the Truth

- Brown et al. (2024) point out that their model-based approach assumes independence over both scientists and clips
 - reasonable assumption for scientists, but there are obvious environmental dynamics that create dependencies over clips
 - “frogs come and frogs go” . . .



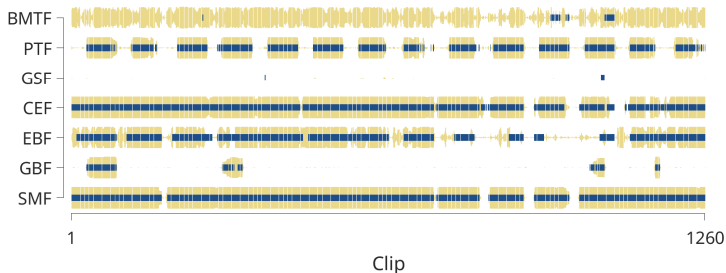
Environment Dynamics Model and the Truth

- The environmental model has a switching probability γ_f for frog f

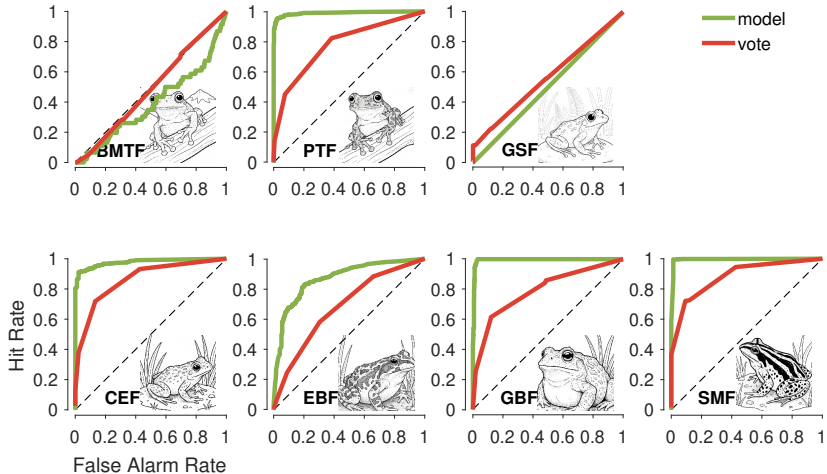
$$\gamma_f \sim \text{Uniform}(0, 1)$$

$$\tau_{f1} \sim \text{Bernoulli}\left(\frac{1}{2}\right)$$

$$\tau_{fk} \sim \begin{cases} \text{Bernoulli}(\gamma_f) & \text{if } \tau_{f(k-1)} = 1 \\ \text{Bernoulli}(1 - \gamma_f) & \text{if } \tau_{f(k-1)} = 0, k > 1 \end{cases}$$



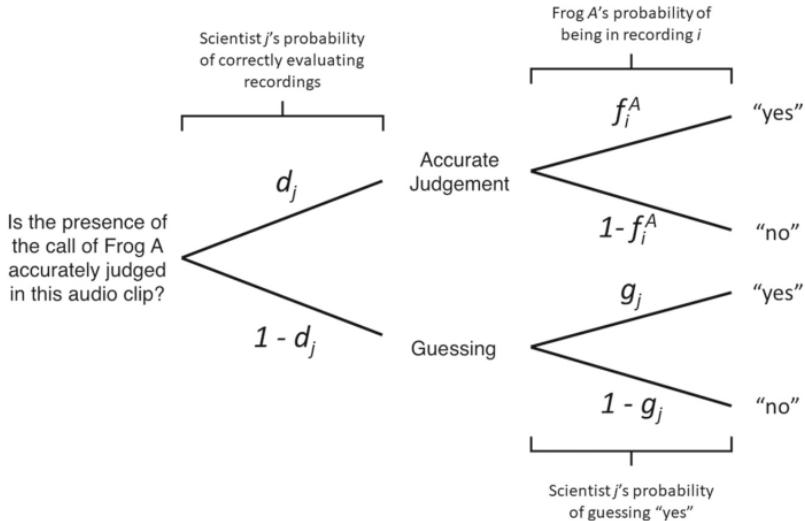
Environment Dynamics Model Performance



References

- Brown, S. D., Sumnall, J., & Smith, P. R. (2024). Using a cognitive model to understand crowdsourced data from citizen scientists. *Behavior Research Methods*, 56(1), 421–434. doi: 10.3758/s13428-023-02289-w
- Lee, M. D. (2024). Using cognitive models to improve the wisdom of the crowd. *Current Directions in Psychological Science*, 33(5), 444–451. doi: 10.1177/09637214241264292

Brown et al. (2024) Model



Brown et al. (2024) Results

